NIMT - CALIBRATION NETWORK LABORATORIES

NIMT-CNL-SP No. 601

Standard Procedure for Calibration of Pressure Measuring Instruments

Type/Model : Hydraulic Type

Issued on January, 2004
First revision on October, 2004
Approved by NIMT

NIMT

HYDRAULIC PRESSURE MEASURING INSTRUMENT

1. Calibration Description:

Parameter	Performance Specifications	Test Method
Gauge Pressure Measurement	Range : 0,1 MPa to 120 MPa (Note 1) Uncertainty : Lower than 0,18 kPa in the range up to 6 MPa Lower than 1,18 kPa in the range up to 120 MPa (Note 2)	Comparison with Standard Hydraulic Pressure Balance

2. Equipment Requirements: Note 2, Note 3, Note 5

Equipment used	Specifications Requirement	Calibration Equipments
1. Pressure Balance, Model AAA with Piston & Cylinder Model : xxx S/N : 111 Model : yyy S/N : 222 Mass set Model : zzz S/N : 333	Range : 0,1 MPa to 6 MPa Uncertainty (k=2) : ± $8{,}0 \times 10^{-5}$*Pe Not lower than 0,18 kPa Range : 2 MPa to 120 MPa Uncertainty (k=2): ± $8{,}0 \times 10^{-5}$*Pe Not lower than 1,18 kPa Range : 50 kg Uncertainty (k=2) : ± 10×10^{-6}*m	1 Piston & Cylinder Model : xxx S/N : 111 2 Piston & Cylinder Model : yyy S/N : 222 3 Mass set Model : zzz S/N : 333
2. Piston & Cylinder Indicator (Note 4) Model : uuu S/N : 444	Uncertainty (k=2, for temperature measurement) : ± 0,5 °C	4 Piston & Cylinder Indicator Mode : uuu S/N : 444
3. Steel Ruler S/N : 555	Uncertainty (k=2) : ± 0,04 mm	5 Steel Ruler S/N : 555
4. Pressure Transmitting (Note 6) Fluid = Hydraulic oil	Density : (914 ± 10) kg/m^3 Surface Tension : (0,031 ± 0,003) N/m	
5. Digital Multimeter Model : mmm S/N : 666	Range : 0 – 100 Vdc Uncertainty : 10 ppm Range : 0 – 100 mAdc Uncertainty : 50 ppm	

3. Preliminary Operation:

3.1 Read the entire procedure before beginning a calibration.

3.2 Visual check the Unit Under Calibration (UUC) to ensure that:

3.2.1 The line voltage required of the UUC has bee n set to 220 VAC, 50 Hz. If the UUC line voltage cannot be selected to be 220 VAC, the proper adapter is required.

IMPORTANT NOTE:

- ***CLEAN GLOVES WITHOUT TALCUM POWDER MUST BE WORN WHILE HANDLING PISTON & CYLINDER OF THE STANDARD PRESSURE BALANCE AND DURING CALIBRATION.***
- ***REFERENCE LEVEL OF THE UUC SHALL BE IDENTIFIED.***

3.3 Connect equipment as shown in figure 1 or figure 2 by using short, clean and appropriate pressure tubes. The reference level of UUC should be the same level as the reference of the standard ± 2 millimeters. If not the different level shall be measured and taken into consideration.

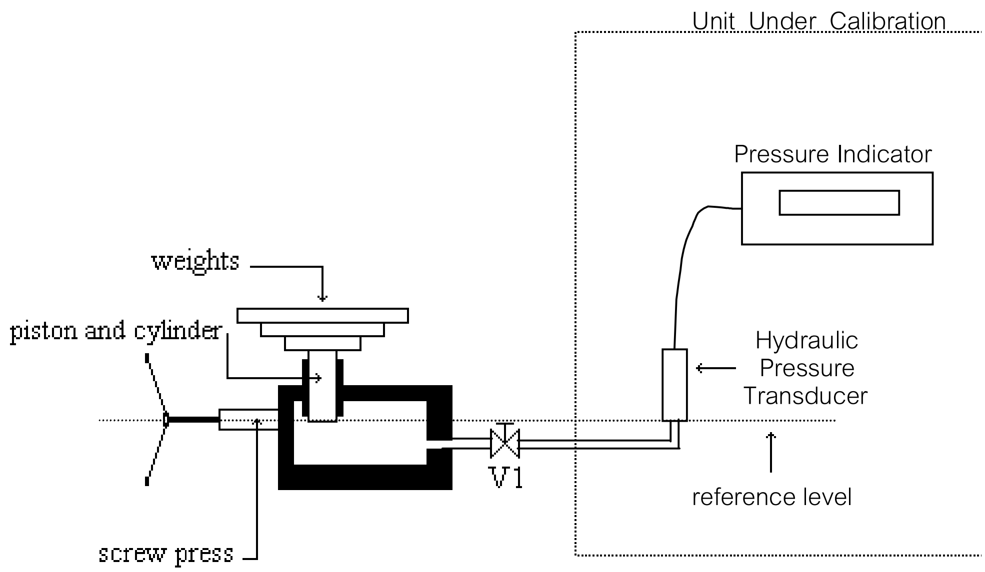

Figure 1

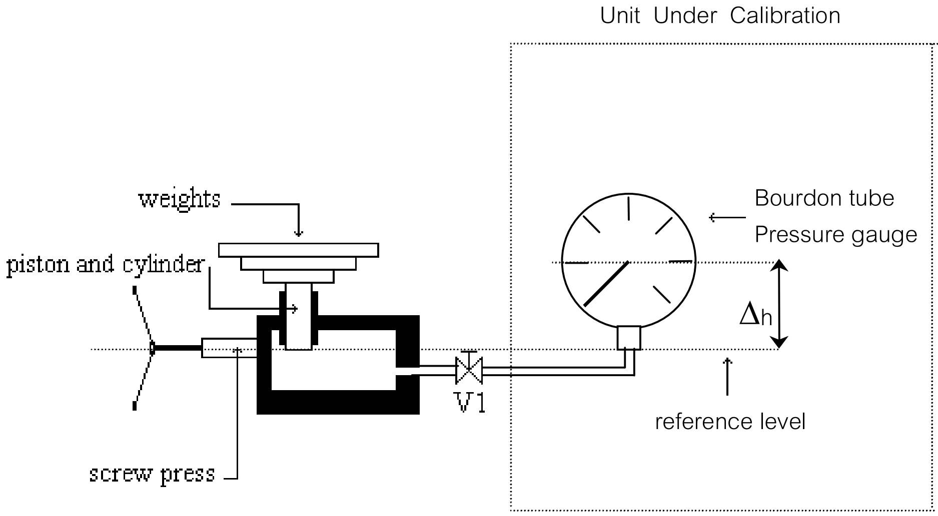

Figure 2

3.4 If the UUC is needed to measure an electrical output, voltage or current ; and the measuring instrument is not provided. It can be calibrated using the laboratory digital multimeter. However it shall be stated in the calibration certificate provided to the customer.

3.5 Care must be taken to ensure that no leakage of pressure between the standard pressure balance and the UUC which will cause some confusion or incorrect reading.

4. Pressure Cycle Exercise:

Unless otherwise specified, the pressure cycles must be applied to exercise (preloading) the UUC according to the accuracy class stated in the DKD R 6-1: 2002, pressure calibration guideline. The pressure cycle must be consisted of starting at zero, going to full scale then back to zero pressure as described in table 1 below.

Table 1: Calibration sequences

Calibration sequence	Measurement uncertainty aimed at, in % of the meas-urement span (*)	Number of meas-urement points with zero up/down	Number of preload-ings	Load change + waiting time (**) seconds	Waiting time at upper limit of measurement range (***) minutes	Number of measurement series up	down
A	< 0,1	9	3	>30	2	2	2
B	0,1 … 0,6	9	2	>30	2	2	1
C	> 0,6	5	1	>30	2	1	1

(*) Reference to the span was used to allow the sequence (necessary calibration effort) to be selected from the table, as the accuracy specifications of the manufacturers are usually related to the measurement span.

(**) One has in any case to wait until steady-state conditions (sufficiently stable indication) of standard and calibration item are reached.

(***) For Bourdon tube pressure gauges, a waiting time of five minutes is to be observed. For quasi-static calibrations (piezoelectric sensor principle), the waiting times can be reduced.

IMPORTANT NOTE:

PRESSURE SURGES CAN DAMAGE THE PRESSURE SENSOR, SMOOTH PRESSURE CHANGES IS REQUIRED THROUGHOUT THE CALIBRATION.

5. Calibration Procedure:

IMPORTANT NOTE:

- ***BEFORE CALIBRATION, A ZERO READING AT THE UUC WHILE THE HYDRAULIC IS OPENED TO AN ATMOSPHERE AFTER EXERCISE SHALL BE OBSERVED. IF NOT, ZERO SET IS NEEDED.***
- ***THE NUMBER OF THE PRESSURE CALIBRATION POINTS OF THE UUC CAN BE DETERMINED ACCORDING TO THE REQUIREMENT IN THE DKD R 6–1 : 2002 CALIBRATION OF PRESSURE GAUGES GUIDELINE WHICH DERCRIBED IN TABLE 1.***

- ***AFTER APPLYING THE INPUT PRESSURE AT EVERY MEASUREMENT POINT, 30 SECONDS MINIMUM ARE NEEDED TO WAIT BEFORE TAKING A READING.***

5.1 Generate a known pressure, step by step; from the minimum to maximum range which explained in table 1 to the UUC. An accurate input pressure which being developed by standard pressure balance can be calculated from the following equation.

$$P_e = \frac{\sum m_i g(1-\rho_a/\rho_{mi}) + V(\rho_f - \rho_a)g + \sigma c}{A_p[1+(\alpha_p+\alpha_c)(t-t_r)]}$$

where :

P_e	is the gauge pressure measure at the bottom of the piston,	Pa.
m_i	is the individual mass value of each weight applied on the piston including all floating elements,	kg.
g	is the local gravity,	m/s^2.
ρ_a	is the density of air,	kg/m^3.
ρ_f	is the density of the measuring fluid,	kg/m^3.
ρ_{mi}	is the density of each weight,	kg/m^3.
V	is the fluid buoyancy correction volume,	m^3.
σ	is the surface tension of the fluid,	N/m.
c	is the circumference of the piston where it emerges from the pressure fluid,	m.
A_p	is the effective area of the piston-cylinder assembly at a reference temperature t_r (20 °C) and at pressure P	m^2.

However, A_p can be expressed from the effective area at null pressure A_o and the first order pressure distortion coefficient.

$$A_p = A_o(1+\lambda p)$$

where :

A_0	=	effective area of the piston & cylinder unit measured at atmospheric pressure and at the reference temperature t_{ref},	m.
λ	=	elastic distortion coefficient of the piston & cylinder unit,	Pa^{-1}.

p	is an approximate value of the measured pressure, It can be calculated as the equation below.	Pa.

$$P = \frac{\Sigma m_i g\left(1 - \frac{\rho a}{\rho m}\right)}{A_0}$$

α_p	is the linear thermal expansion coefficient of the piston,	K^{-1}.
α_c	is the linear thermal expansion coefficient of the cylinder,	K^{-1}.
t	is the measured temperature of the piston-cyli nder assembly during its use,	^{o}C.

When the pressure p_m is expressed at the reference level of the UUC which may be different from reference level of the standard. A corrective term (head correction) has to be added to the pressure *Pe*; expressed above.

p_m	=	$p_e + (\rho_f - \rho_a)g\Delta h$	
p_m	=	the pressure measurand at the UUC's reference-level,	Pa.

where:
Δh is the difference between the altitude h_1 of the balance reference level and the altitude h_2 of the point where the pressure has to be measured, m.

$$\Delta h = h_1 - h_2$$

5.2 When the standard pressure balance is obtained at mid float position, the UUC reading has to be observed and recorded in the table attached in appendix A.

IMPORTANT NOTE:

DURING THE CALIBRATION, THE VALVE V_1 MAY BE USED TO PREVENT HYSTERESIS WHICH MAY BE OCCURED ON THE UUC WHILE CHANGING MASSES OF THE STANDARD PRESSURE BALANCE.

5.3 Repeat step 5.1 and 5.2 until the calibration is completed.

5.4 Release all pressure from the measurement system and disconnect all of the equipments.

6. Data Analysis and Uncertainty Calculation

The measurement result and uncertainty of measurement can be calculated by using the method stated in DKD R 6-1: 2002, Calibration of Pressure Gauges.

APPENDIX A

Client: ....

Unit: .... Type: .... No.: .... Date: ....

Range: ....

Resolution: .... Calibration cycle: .... ____________________

Accuracy: .... Name of Calibrator

Pressure Calibration Table

p in	t_{room} in C	p_{amb} in bar	t_{piston} in C	trim mass in g	plate no.		UUC reading in.........

Standard - system	
Standard-mass-set	
Height difference	cm (+when std. is higher)
Pressure medium	Calibrator signature

APPENDIX B

References:

1. BS EN 837-1: 1998 British Standard "Pressure Gauge Part 1. Bourdon tube pressure gauges-dimensions, metrology, requirements and testing".
2. DKD R 6-1: 2002 "Calibration of Pressure Gauges" published by the Physikalisch
   – Technische Bundesanstalt (PTB) in cooperation with the Technical Committee "Pressure" of the Deutscher Kalibrierdienst (DKD), Braunschweig 2002.
3. EAL-G26 "calibration of Pressure Balance" European cooperation for Accreditation of Laboratories: Edition 1, July 1997.
4. Guide to the Expression of Uncertainty in Measurement 1 st edition 1993, ISO, Geneva, ISBN 92-67-10188-9.

Note:

1. The pressure range is depended on customer's standard equipment.
2. The uncertainty values are for example. The laboratories shall evaluate by themself.
3. The model and serial numbers are for ex ample.
4. If available in the laboratory.
5. The standard equipments can be added based on each laboratory.
6. The pressure transmitting fluid property is based on the fluid used in each laboratory.

Attachment 1

Mathematical model:

$$\Delta P = P_{ind} - P_{std} + \Sigma\delta P_i$$

$$\Sigma\delta P_i = \delta P_{std} + \delta P_{resolution} + \delta P_{zerodeviation} + \delta P_{repeatability} + \delta P_{hysteresis} + \delta P_{\Delta h}$$

$$U = k \cdot \sqrt{u^2_{s\tan dard} + u^2_{resolution} + u^2_{zerodeviation} + u^2_{repeatability} + u^2_{hysteresis} + u^2_{\Delta h}}$$

Where:

P_{ind}	=	UUC reading,	Pa
P_{std}	=	Pressure standard at the the point to be measured,	Pa
δP_{std}	=	The correction value of pres sure standard,	Pa
$\delta P_{resolution}$	=	The correction value due to resolution of the UUC reading,	Pa
$\delta P_{zero\ deviation}$	=	The correction value due to the UUC zero reading,	Pa
$\delta P_{repeatability}$	=	The correction value due to the UUC reading repeatability,	Pa
$\delta P_{hysteresis}$	=	The correction value due to hysteresis of the UUC reading,	Pa
$\delta P_{(\Delta h)}$	=	The correction value due to the head correction measurement system,	Pa

Since pressure balance is used as a standard equipment.

Therefore:

$$\delta P_{std} = \delta P_{certificate} + \delta P_{temperature} + \delta P_{gravity} + \delta P_{(\rho a)}$$

$$u^2_{std} = u^2_{certificate} + u^2_{temperature} + u^2_{gravity} + u^2_{\rho a}$$

Where:

$\delta P_{certificate}$	=	The correction value due to the calibration certificate of the standard hydraulic pressure balance,	Pa
$\delta P_{temperature}$	=	The correction value of the piston & cylinder temperature measurement,	Pa
$\delta P_{gravity}$	=	The correction value due to local gravity acceleration,	Pa
$\delta P_{(\rho a)}$	=	The correction value due to an ambient air density measurement,	Pa

Note:

$$u_{temperature} = -(\alpha+\beta) \bullet p \bullet \sqrt{\frac{1}{3} \cdot a_t^2} \qquad u_{\Delta h} = (\Delta p \bullet g) \bullet \sqrt{\frac{1}{3} \cdot a_h^2}$$

$$u_{gravity} = \left(\frac{p}{g}\right) \cdot \sqrt{\frac{1}{3} \cdot a_g^2} \qquad u_{\rho a} = -\left(\frac{p}{\rho_m}\right) \cdot \sqrt{\frac{1}{3} \cdot a_{\rho_a}^2}$$

Where:

$a_t, a_h, a_g, a_{\rho a}$ = Uncertainty attributed to the input quantity

<u>Example</u>

Uncertainty budget for the calibration of a hydraulic pressure transducer

Calibration effort for calibration sequence A

Statement of mean value (*Miw*) with measurement deviation (Δp), repeatability (*b*) and hysteresis (*h*)

<u>Calibration item</u>

Accuracy stated by manufacturer	:	0,04 %Rdg + 0,005 %FS
Measurement range	:	*200 bar*
Resolution	:	0,01 bar

<u>Standard device</u>

Expanded uncertainty (standard)	:	0,008 %rdg but not smaller than 0,000 4 bar
Linear thermal expansion coefficient ($\alpha_c+\alpha_p$)	:	*0,000 009* K^{-1}
Temperature of piston (t_{piston})	:	(20 ± 0,5)°C
Density of mass (ρ_m)	:	7 800 kg/m^3

<u>Calibration conditions</u>

Pressure-transmitting medium	:	Oil
Fluid density ($\rho_{f,20°C, 1 bar}$)	:	(914 ± 10) kg/m^3
Air density ($\rho_{a,20°C, 1 bar}$)	:	(1,199 ± 0,015) kg/m^3
Height different (Δh)	:	(0 ± 0,002) m
Gravity (g)	:	(9,782 970 ± 0,000 4 9) m/s^2
Ambient temperature	:	(20 ± 1)°C
Ambient pressure	:	(1 013 ± 10) mbar
Ambient humidity	:	*(55 ± 10) %RH*

***Table 2 : Measurement data**

P **standard** (bar)	**Reading from calibration item** *Pind* (bar)			
	M1	**M2**	**M3**	**M4**
0,000	0,00	0,00	0,00	0,00
20,022	19,99	19,99	19,97	19,98
40,030	39,97	39,98	39,95	39,97
59,970	59,87	59,90	59,86	59,88
79,979	79,88	79,90	79,86	79,88
99,988	99,88	99,89	99,85	99,87
119,997	119,87	119,89	119,85	119,87
140,005	139,87	139,89	139,85	139,87
160,014	159,88	159,91	159,87	159,89
180,022	179,87	179,91	179,87	179,88
200,031	199,90	199,91	199,88	199,90

***Note:**

- All value are presented after normalization.

Table 3: Evaluation

Mean value *Miw* $\sum M_i/4$	**Deviation** Δp $Miw\text{-}p_e$	**Repeatability** *b* *Max(|M3-M1|, |M4-M2|)*	**Hysteresis** *h* *(|M2-M1|+ |M4-M3|)/2*	**Expanded uncertainty** *U*
bar	bar	bar	bar	bar
0,00	0,00	0,00	0,00	0,013
19,98	-0,04	0,02	0,01	0,018
39,97	-0.06	0,02	0,02	0,020
59,88	-0,09	0,02	0,03	0,023
79,88	-0,10	0,02	0,02	0,022
99,87	-0,12	0,03	0,02	0,025
119,87	-0,13	0,02	0,02	0,023
139,87	-0,14	0,02	0,02	0,024
159,89	-0,12	0,02	0,02	0,026
179,88	-0,14	0,03	0,03	0,030
199,90	-0,13	0,02	0,02	0,026

Uncertainties distributed

$\delta P_{standard,\ cer}$

Width of distribution (k = 2)	=	$\left(\frac{0{,}008}{100} * 200{,}031\right)$	bar
	=	$1{,}60*10^{-2}$	bar
Probability distribution	=	Normal	
Uncertainty, $u(x_i)$	=	$\frac{1{,}60*10^{-2}}{2}$	bar
	=	$8{,}00*10^{-3}$	bar
Sensitivity coefficient, (c_i)	=	1	
∴ Uncertainty contribution, $u(y)$	=	$(c_i)*\ u(x_i)$	
	=	$8{,}00*10^{-3}$	bar

$\delta P_{temperature}$

Width of distribution (2a)	=	$\pm\ 0{,}5$	K
Probability distribution	=	Rectangular	
Uncertainty, $u(x_i)$	=	$\frac{0{,}5}{\sqrt{3}}$	K
	=	$2{,}89*10^{-1}$	K
Sensitivity coefficient, (c_i)	=	$-(\alpha+\beta)*p$	
	=	$-9*10^{-6}\ K^{-1}*200{,}031$	bar
	=	$-1{,}80*10^{-3}$	bar/K
∴ Uncertainty contribution, $u(y)$	=	$(c_i)*\ u(x_i)$	
	=	$-1{,}80*10^{-3}$ bar/K$*2{,}89*10^{-1}$	K
	=	$-5{,}20*10^{-4}$	bar

$\delta P_{gravity}$

Width of distribution (2a)	=	$\pm$ 0,000 49	m s^{-2}
Probability distribution	=	Rectangular	
Uncertainty, $u(x_i)$	=	$\frac{4{,}9*10^{-4}}{\sqrt{3}}$	m s^{-2}
	=	2,83*10^{-4}	m s^{-2}
Sensitivity coefficient, (c_i)	=	$\frac{p}{g}$	
	=	$\frac{200{,}031}{9{,}782\ 97}$	bar/ m s^{-2}
	=	20,45	bar/ m s^{-2}
$\therefore$ Uncertainty contribution, $u(y)$	=	$(c_i)* u(x_i)$	
	=	20,45 bar/ m$\cdot$s^{-2} * 2,83*10^{-4} m s^{-2}	
	=	5,79*10^{-3}	bar

$\delta P_{(\rho a)}$

Width of distribution (2a)	=	$\pm$ 0,015	kg m^{-3}
Probability distribution	=	Rectangular	
Uncertainty, $u(x_i)$	=	$\frac{0{,}015}{\sqrt{3}}$	kg m^{-3}
	=	8,66*10^{-3}	kg m^{-3}
Sensitivity coefficient, (c_i)	=	$-\frac{p}{\rho_m}$	
	=	$-\frac{200{,}031}{7\ 800}$	bar/ kg m^{-3}
	=	-2,56*10^{-2}	bar/ kg m^{-3}
$\therefore$ Uncertainty contribution, $u(y)$	=	$(c_i)* u(x_i)$	
	=	-2,56*10^{-2} bar/ kg$\cdot$m^{-3} *8,66*10^{-3} kg$\cdot$m^{-3}	
	=	-2,22*10^{-4}	bar

$\delta P_{(\Delta h)}$

Width of distribution (2a)	=	$\pm$ 0,002	m
Probability distribution	=	Rectangular	
Uncertainty, $u(x_i)$	=	$\frac{0{,}002}{\sqrt{3}}$	m
	=	$1{,}15 * 10^{-3}$	m

$$\rho_{f(p,t)} = \rho_{f(20^{\circ}C,\ 1bar)} \left(1 - 0{,}000\ 78\left[\left(\frac{t_{room}(^{\circ}C) + t_{piston}(^{\circ}C)}{2}\right) - 20\right]\right)(1+0{,}000\ 08P(bar)) \quad (1)$$

$$= 914 \left(1 - 0{,}000\ 78\left[\left(\frac{20+20}{2}\right) - 20\right]\right)(1+0{,}000\ 08 * 200{,}031)$$

$$= 928{,}63 \text{ kg m}^{-3}$$

Note : Equation (1) reference from PTB

Sensitivity coefficient, (c_i)	=	$\Delta\rho * g$	
	=	$(928{,}63-1{,}199)$ kg m^{-3} * 9,782 97 m s^{-2}	
	=	$9{,}07 * 10^{-2}$	bar/m
$\therefore$ Uncertainty contribution, $u(y)$	=	$(c_i) * u(x_i)$	
	=	$9{,}07 * 10^{-2}$ bar/m * $1{,}15 * 10^{-3}$	m
	=	$1{,}04 * 10^{-4}$	bar

$\delta P_{resolution}$

Width of distribution (2a)	=	0,01	bar
Probability distribution	=	Rectangular	
Uncertainty, $u(x_i)$	=	$\frac{0{,}01}{2\sqrt{3}}$	bar
	=	$2{,}89 * 10^{-3}$	bar
Sensitivity coefficient, (c_i)	=	1	
$\therefore$ Uncertainty contribution, $u(y)$	=	$(c_i) * u(x_i)$	
	=	$2{,}89 * 10^{-3}$	bar

$\delta P_{zero\ deviation}$

Uncertainty contribution, $u(y)$ = 0,00 bar

$\delta P_{repeatability}$

Width of distribution (2a) = 0,02 bar

Probability distribution = Rectangular

Uncertainty, $u(x_i)$ = $\dfrac{0{,}02}{2\sqrt{3}}$ bar

= $5{,}77 \cdot 10^{-3}$ bar

Sensitivity coefficient, (c_i) = 1

$\therefore$ Uncertainty contribution, $u(y)$ = $(c_i) * u(x_i)$

= $5{,}77 \cdot 10^{-3}$ bar

$\delta P_{hysteresis}$

Width of distribution (2a) = 0,02 bar

Probability distribution = Rectangular

Uncertainty, $u(x_i)$ = $\dfrac{0{,}02}{2\sqrt{3}}$ bar

= $5{,}77 \cdot 10^{-3}$ bar

Sensitivity coefficient, (c_i) = 1

$\therefore$ Uncertainty contribution, $u(y)$ = $(c_i) * u(x_i)$

= $5{,}77 \cdot 10^{-3}$ bar

Expanded uncertainty $= k \cdot \sqrt{u_{cer}^2 + u_{temp}^2 + u_g^2 + u_{\rho a}^2 + u_{\Delta h}^2 + u_{res}^2 + u_{zero}^2 + u_{repeat}^2 + u_{hys}^2}$

= 2 * 0,013

= 0,026 bar

Table 4: Uncertainty budget for load step p = 200,031 bar

Quantity X	Estimate x_i	Width of distribution $2a$	**Divisor**	Uncertainty $u(x_i)$	Sensitivity coefficient c_i	Uncertainty contribution $u(y)$	Variance u^2
						bar	bar^2
P standard	200,031 bar	$1{,}60*10^{-2}$ bar	2	$8{,}00*10^{-3}$ bar	1	$8{,}00*10^{-3}$	$6{,}40*10^{-5}$
δP standard, temperature		0,5 K	√3	$2{,}89*10^{-1}$ K	$-1{,}80*10^{-3}$ bar/K	$-5{,}20*10^{-4}$	$2{,}70*10^{-7}$
δP gravity		$4{,}90*10^{-4}$ m/s^2	√3	$2{,}83*10^{-4}$ m/s^2	20,45 bar/ms^{-2}	$5{,}79*10^{-3}$	$3{,}35*10^{-5}$
δP (ρ_a)		0,015 kg/m^3	√3	$8{,}66*10^{-3}$ kg/m^3	$-2{,}56*10^{-2}$ bar/kg m^3	$-2{,}22*10^{-4}$	$4{,}93*10^{-8}$
***δP height*		0,002 m	√3	$1{,}15*10^{-3}$ m	$9{,}07*10^{-2}$ bar/m	$1{,}04*10^{-4}$	$1{,}08*10^{-8}$
P ind, resolution	199,90 bar	0,01 bar	2√3	$2{,}89*10^{-3}$ bar	1	$2{,}89*10^{-3}$	$8{,}35*10^{-6}$
δP zero deviation		0,00 bar	2√3	0,00 bar	1	0,00	0,00
δP repeatability		0,02 bar	2√3	$5{,}77*10^{-3}$ bar	1	$5{,}77*10^{-3}$	$3{,}33*10^{-5}$
δP hysteresis		0,02 bar	2√3	$5{,}77*10^{-3}$ bar	1	$5{,}77*10^{-3}$	$3{,}33*10^{-5}$
ΔP	-0,13 bar			u =		0,013	$\sum u_i^2 = 1{,}73*10^{-4}$
ΔP	-0,13 bar	U = k.u		(k = 2)		**0,026 bar**	

**allowing for the pressure-dependent fluid density